

Crime in London

Exploratory Analysis of 2023 Crime Data



Use Case

Context

I came across a source of crime data for London last year while working on an assignment for the London School of Economics. For that assignment I was interested in cycle theft, but having seen the data that was available it piqued my interest in overall.

I was particularly interested in exploring the data to understand:

- *Crime rate over time for*
- *Which crimes occur most frequently and where do they occur*
- *Which London boroughs have the highest and lowest crime rates*

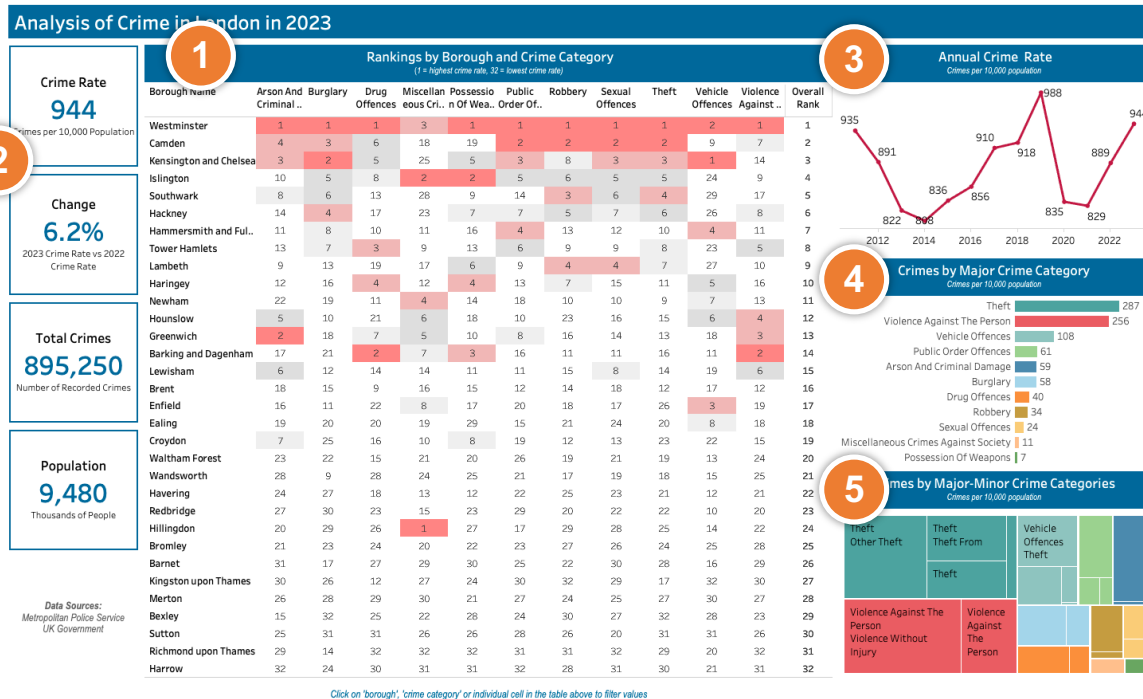
The key metric I chose to use was the number of crimes per 10,000 population which creates a standardised measure and removes the impact of population size on the number of crimes.

Proposed Solution:

Create a Tableau dashboard that allows the user to investigate:

- A. Is the crime rate in London increasing?
- B. Which crimes occur most frequently and where?
- C. Which London boroughs have the lowest and highest crime rates?
- D. What else does that data show that is worthy of further investigation?

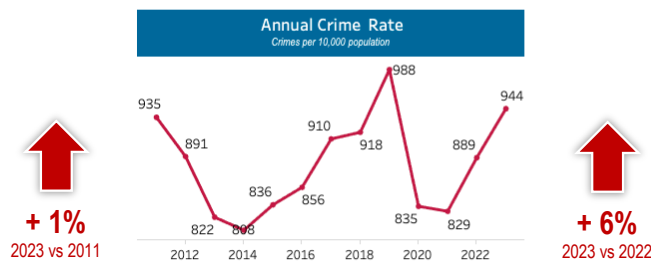
Dashboard Overview



- 1 Interactive heatmap showing the rank of each borough by major crime category and overall rank in 2023 (1 = highest crime rate, 32 = lowest crime rate). The 10 boroughs with the highest crime rates are coloured grey to red, red being the highest crime rate. Clicking on a specific 'borough', 'crime category' or individual cell filters the data and updates the other charts and elements on the dashboard.
- 2 Key metrics for London as a whole or the filtered data.
- 3 Line chart showing the trend in the crime rate between 2011 and 2023 for London as a whole or the filtered data.
- 4 Bar chart showing the the most frequently occurring crimes in 2023 in London as a whole or the filtered data.
- 5 Tree map showing proportion of crimes by major and minor crime category in 2023 in London as a whole or the filtered data.

Is the crime rate in London increasing?

Overall crime rate



Overall, the 2023 crime rate is 1% higher than 2011 and 6% higher than prior year (2022).

The crime rate decreased between 2011 and 2014 by 14% to its' lowest level in the study at 808 crimes per 10,000 population. It then rose to its' peak year of 988 crimes per 10,000 population in 2019.

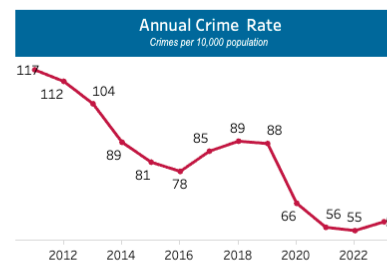
The COVID-19 pandemic in 2020-21 led to a reduction in crime rate, but since the end of the pandemic it has risen to 944 crime per 10,000 population (the second highest during the study period).

Long-term trends by major crime category 2011-23

Burglary



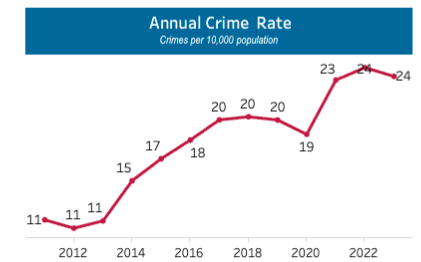
51%
2023 vs 2011



Sexual Offences



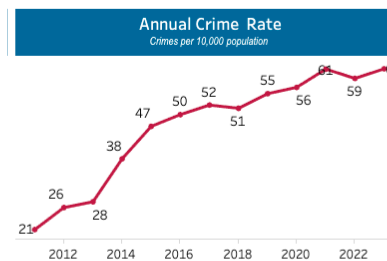
+ 218%
2023 vs 2011



Public Order Offences



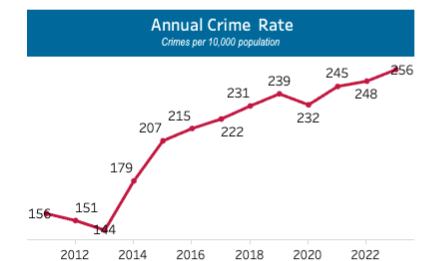
+ 290%
2023 vs 2011



Violence Against The Person

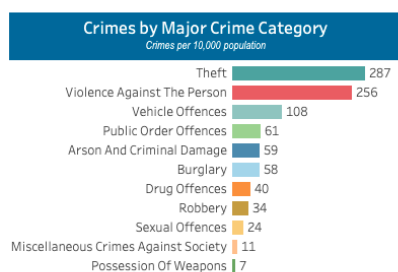


+ 164%
2023 vs 2011



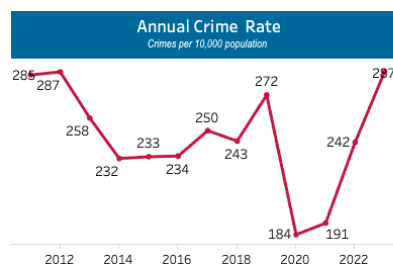
Which crimes occur most frequently and where?

2023 Crime by Category



The two most frequently occurring crimes in 2023 were Theft (287 crimes per 10,000 population) and Crimes Against the Person (256 crimes per 10,000 population). These two crimes account for 58% of all recorded crimes in London.

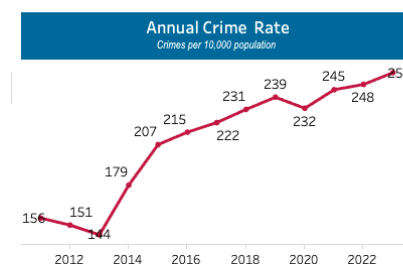
Theft



Worst 5 boroughs

1. Westminster
2. Camden
3. Kensington and Chelsea
4. Southwark
5. Islington

Violence Against the Person



Worst 5 boroughs

1. Westminster
2. Dagenham and Barking
3. Greenwich
4. Hounslow
5. Tower Hamlets

Which London boroughs have the highest and lowest crime rates?

Westminster
has the highest crime rate

**3,357 crimes per 10,000
population**



Rankings by Borough and Crime Category (1 = highest crime rate, 32 = lowest crime rate)												
Borough Name	Arson And Criminal ..	Burglary	Drug Offences	Miscellan eous Cri..	Possessio n Of Wea..	Public Order Of..	Robbery	Sexual Offences	Theft	Violence Against ..	Vehicle Offences	Overall Rank
Westminster	1	1	1	3	1	1	1	1	1	1	2	1
Camden	4	3	6	18	19	2	2	2	2	7	9	2
Kensington and Chelsea	3	2	5	25	5	3	8	3	3	14	1	3
Islington	10	5	8	2	2	5	6	5	5	9	24	4
Southwark	8	6	13	28	9	14	3	6	4	17	29	5

Harrow
has the lowest crime rate

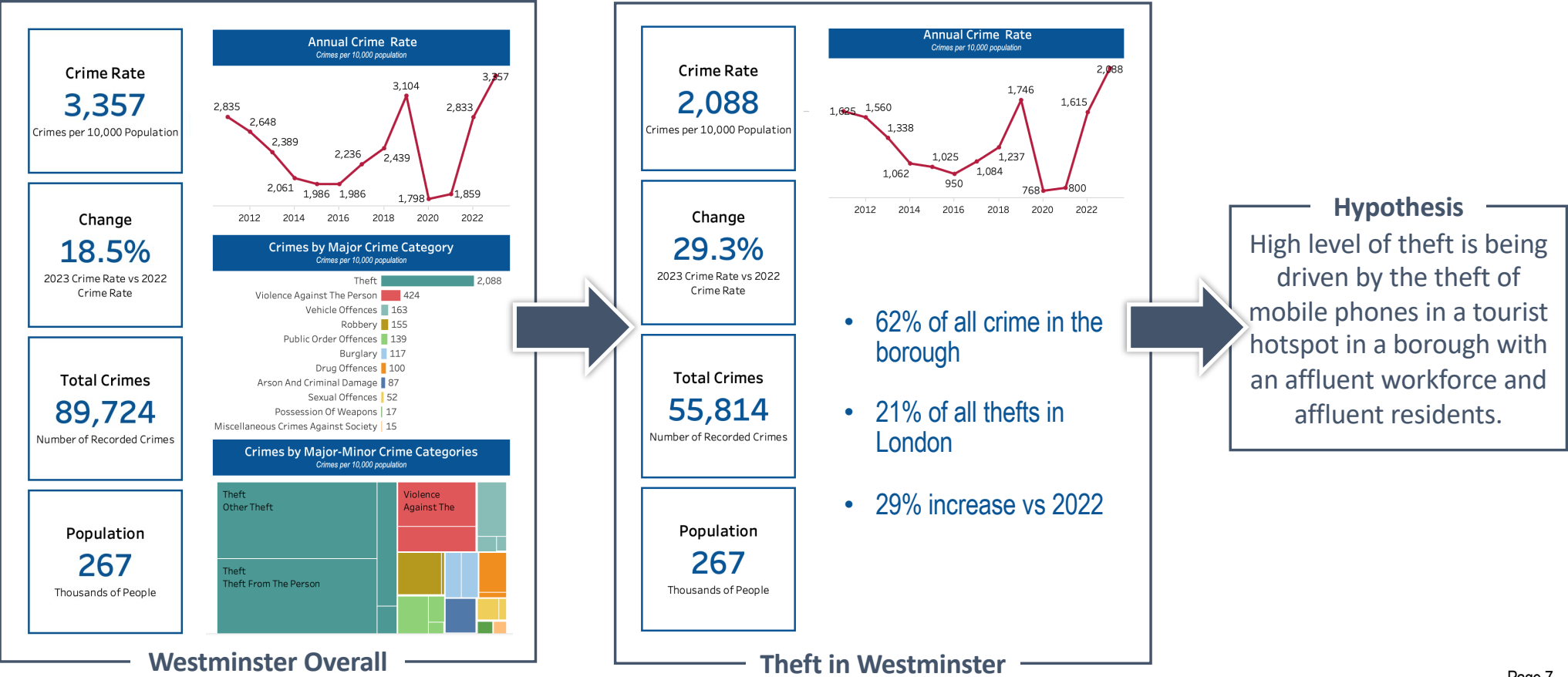
**578 crimes per 10,000
population**



Rankings by Borough and Crime Category (1 = highest crime rate, 32 = lowest crime rate)												
Borough Name	Arson And Criminal ..	Burglary	Drug Offences	Miscellan eous Cri..	Possessio n Of Wea..	Public Order Of..	Robbery	Sexual Offences	Theft	Violence Against ..	Vehicle Offences	Overall Rank
Harrow	32	24	30	31	31	32	28	31	30	31	21	32
Richmond upon Thames	29	14	32	32	32	31	31	32	29	32	20	31
Sutton	25	31	31	26	26	28	26	20	31	26	31	30
Bexley	15	32	25	22	28	24	30	27	32	23	28	29
Merton	26	28	29	30	21	27	24	25	27	27	30	28

What else does that data show that is worthy of further investigation?

Theft in Westminster accounts for 62% of recorded crime in the borough and 21% of all thefts in Greater London



Summary of Findings

- Overall, the 2023 crime rate is 1% higher than 2011 and 6% higher than prior year (2022).
- The two most frequently occurring crimes in 2023 were Theft (287 crimes per 10,000 population) and Crimes Against the Person (256 crimes per 10,000 population). These two crimes account for 58% of all recorded crimes in London.
- Westminster has the highest crime rate with 3,357 crimes per 10,000 population. Harrow has the lowest crime rate with 578 crimes per 10,000 population.
- Theft in Westminster accounts for 62% of recorded crime in the borough and 21% of all thefts in Greater London.
- Westminster has the highest crime rate in Greater London in 9 of the 11 major crime categories and is 2nd and 3rd highest the remaining 2 categories.

Recommendations

- Conduct a deeper dive on Westminster to understand why it has the highest overall crime rate and has the highest crime rate in 9 of the 11 crime categories and is 2nd and 3rd highest the remaining 2 categories.
 - Are there specific locations or hot spots where crimes occur?
 - Are there patterns to when crimes are committed; over time, seasons, days of week, night or day, etc?
 - Are there correlations between the demographics of the victims of crime and the types of crime they are victims of? Are tourists targeted? Is the perception of the borough being affluent and having an affluent work force making the borough a target for crime?
 - What are the demographics of the perpetrators of crime? Do they live inside the borough, other parts of London or outside of London?
 - Are there machine learning models that could identify correlations and cause and effect relationships between different independent factors and crime rates and crime categories?